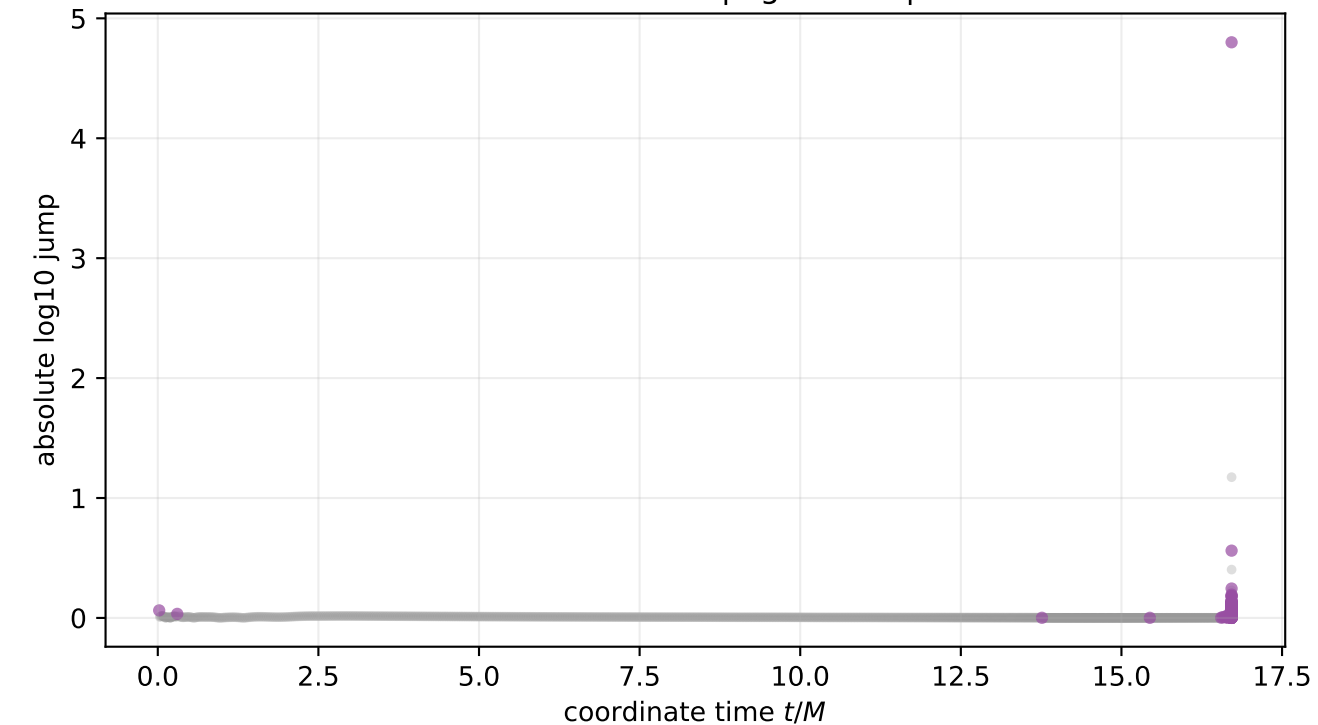
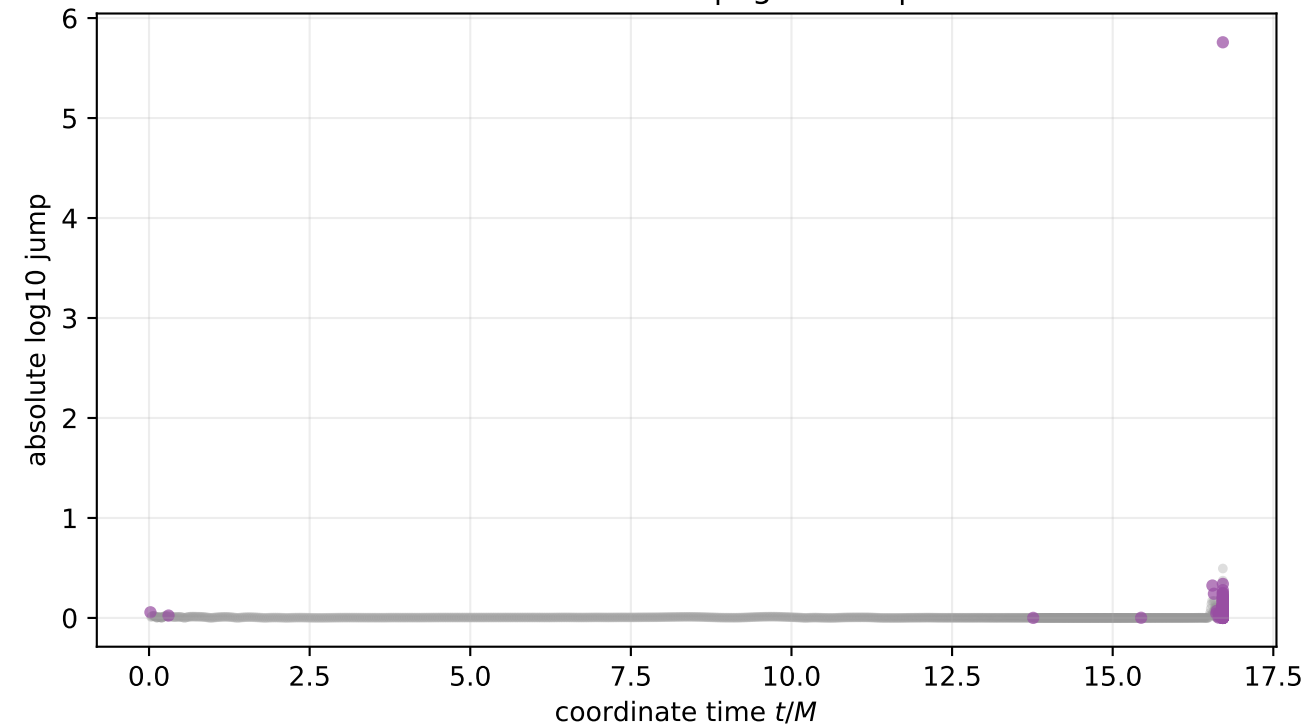


O4 history constraint jumps versus recorded topology changes

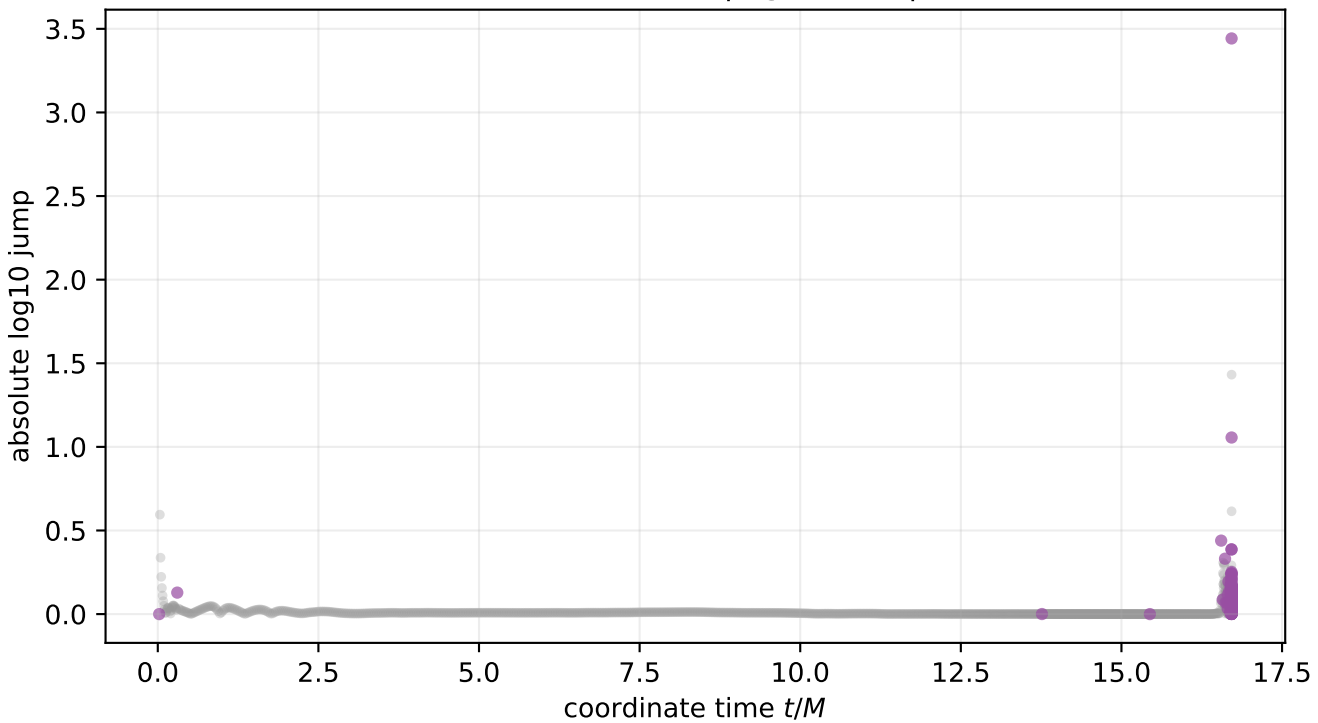
C: consecutive $|\log_{10} \text{ratio}|$



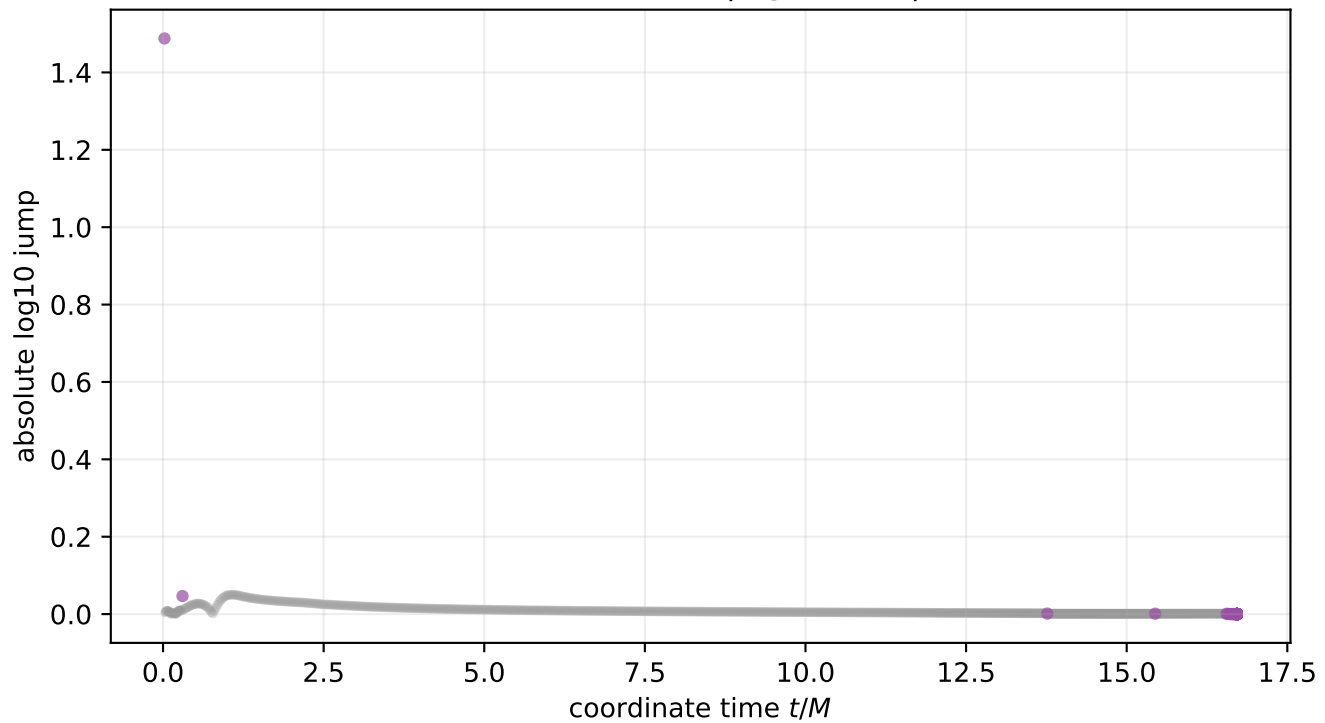
H: consecutive $|\log_{10} \text{ratio}|$



M: consecutive $|\log_{10} \text{ratio}|$



Z: consecutive $|\log_{10} \text{ratio}|$



• same hierarchy • hierarchy changed